Structural dim = 2 for $(3, 3, 1^q)$: outer-factor injectivity via adjacent disjointness

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Abstract

The same-row-dies paper of 13 May night (`2026-05-13-same-row-dies-331.tex`, SRD-331) proved that for $\lambda = (3, 3, 1^q)$, $q \geq 2$,

$$B^{(\lambda)}_{\ell+1} V_\lambda \;=\; (T_\ell{+}1)(T_{\ell+1}{+}1)(T_{n-2}{+}1)\, \Phi_* \Big(B^{(\mu_1)}_{\ell(\mu_1)+1} V_{\mu_1} \Big)$$

with $\mu_1 = (3, 2, 1^{q-1})$ on $n - 2 = q + 4$ letters. Combined with the May-12 evening theorem $\dim B^{(\mu_1)} = 2$ ($q \geq 4$), this gives the upper bound $\dim B^{(\lambda)} \leq 2$ structurally at $q \geq 4$. The matching structural lower bound $\dim B^{(\lambda)} \geq 2$ was open (SRD-331 Remark 4.6); only computational verification at $q \in \{2, 3, 4, 5\}$ was available.

Theorem. For $\lambda = (3, 3, 1^q)$ with $q \geq 4$,

$$\dim B^{(\lambda)}_{\ell+1} V_\lambda \;=\; 2.$$

Proof technique. The new ingredient is a clean injectivity argument for the outer factors. Each factor $(T_j{+}1)$ maps $E^+_{j+1}|_{V_\lambda}$ *injectively* into $E^+_j|_{V_\lambda}$, because (i) its image lies in E^+_j by the quadratic relation $(T_j{-}q)(T_j{+}1) = 0$, and (ii) its kernel E^-_j intersects E^+_{j+1} trivially by the May-19 abstract adjacent-disjointness lemma (in its symmetrically swapped form). Chaining the three factors gives an injective composition $E^+_{n-1} \to E^+_{n-2} \to E^+_{n-3} \to E^+_{n-4}$. Since $\Phi_*(B^{(\mu_1)}) \subseteq E^+_{n-1}$ has dim 2, its image has dim 2, giving the lower bound.

What this generalises. The proof works for any $r = 1$ rank-zero shape λ where (a) the SRD reduction formula applies (same-row-dies holds and the corner-pair 2-block is non-degenerate), and (b) $\dim B^{(\mu_1)}$ is known. The upper-and-lower bound then collapses to the equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$. Specialising to $\lambda = (3, 3, 1^q)$ with $\mu_1 = (3, 2, 1^{q-1})$ and May-12 evening's dim $= 2$ gives this paper's main theorem.

Computational verification. The injectivity chain $E^-_j \cap E^+_{j+1} = 0$ verified for $j \in \{n - 4, n - 3, n - 2\}$ and the outer- chain rank on E^+_{n-1} equals $\dim E^+_{n-1}$ at $q \in \{4, 5, 6\}$.

1 Setup and statement

Notation follows the SRD-331 paper. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$; V_λ is the seminormal cell module for $\lambda \vdash n$; $E^\pm_j$ are the q- and (-1)-eigenspaces of T_j on V_λ;

$$R'_k \;:=\; (T_{k-1}{+}1) \cdots (T_1{+}1), \qquad B^{(\lambda)}_j V_\lambda \;:=\; R'_j \cdots R'_n V_\lambda.$$

Throughout this paper, $\lambda := (3, 3, 1^q)$ with $q \geq 4$, $n := q{+}6$, $\ell := q{+}2$, $j_0 := \ell{+}1 = q{+}3 = n{-}3$. The corners of λ are $c_1 := (2, 3)$ (length-preserving) and $c_* := (q + 2, 1)$ (column-1 bottom). Let $\mu_1 := \lambda \setminus \{c_1, c_*\} = (3, 2, 1^{q-1}) \vdash n - 2$, which has $\ell(\mu_1) = q + 1$ and $|\mu_1| = q + 4 \geq 8$ when $q \geq 4$. The axial distance of the corner-pair 2-block is $d := c(c_*) - c(c_1) = (1 - (q{+}2)) - (3 - 2) = -(q{+}2)$, so $|d| \geq 6$ at $q \geq 4$ (in particular non-degenerate).

The branching iso Φ_*. The pair $\{c_1, c_*\}$ gives an $H_q(S_{n-2})$-equivariant injection $\Phi_* : V_{\mu_1}|_{n-2} \hookrightarrow E^+_{n-1}|_{V_\lambda}$ (SRD-331 Lemma 3.2 / r=1 paper Lemma 2.4). For each SYT $T' \in \mathrm{SYT}(\mu_1)$, $\Phi_*(v_{T'})$ is the normalised $+q$- eigenvector of T_{n-1} on the $\{T_A, T_B\}$ 2-block, where T_A puts n at c_*, $n-1$ at c_1, and T_B swaps. Φ_* is T_j-equivariant for $j \le n-3$ (index gap ≥ 2 from T_{n-1}).

Recall: SRD-331 reduction formula.

Theorem 1 (SRD-331 Corollary 4.3)**.** *For $\lambda = (3, 3, 1^q)$, $q \ge 2$,*

$$B^{(\lambda)}_{\ell+1} V_\lambda \;=\; (T_\ell{+}1)(T_{\ell+1}{+}1)(T_{n-2}{+}1)\,\Phi_*\Big(B^{(\mu_1)}_{\ell(\mu_1)+1} V_{\mu_1}\Big)\,. \tag{1}$$

Equivalently, with the three outer indices $\ell = n-4$, $\ell+1 = n-3$, $n-2$:

$$B^{(\lambda)} \;=\; \mathcal{P}\,\Phi_*(B^{(\mu_1)}), \qquad \mathcal{P} := (T_{n-4}{+}1)(T_{n-3}{+}1)(T_{n-2}{+}1).$$

Recall: the μ_1-dim.

Theorem 2 (May-12 evening Theorem 5.4)**.** *For $\mu_1 = (3, 2, 1^{q-1})$ with $|\mu_1| = q + 4 \ge 8$ (i.e., $q \ge 4$),*

$$\dim B^{(\mu_1)}_{\ell(\mu_1)+1} V_{\mu_1} \;=\; 2.$$

Main result.

Theorem 3 (Main)**.** *For $\lambda = (3, 3, 1^q)$ with $q \ge 4$,*

$$\dim B^{(\lambda)}_{\ell+1} V_\lambda \;=\; 2. \tag{2}$$

By Theorem 1, $B^{(\lambda)} = \mathcal{P}\,\Phi_*(B^{(\mu_1)})$. Hence

$$\dim B^{(\lambda)} \;\le\; \dim \Phi_*(B^{(\mu_1)}) \;=\; \dim B^{(\mu_1)} \;=\; 2$$

(using Φ_* injective by SRD-331 Lemma 2.4 and Theorem 2). This is the upper bound, already in SRD-331. The new content is the matching lower bound, which we prove in §3 via outer-factor injectivity.

2 The two ingredients: image landing and adjacent disjointness

2.1 Image of $(T_j + 1)$ lands in E^+_j

Lemma 4 (Image landing)**.** *For every $H_q(S_n)$-module V and every $1 \le j \le n-1$,*

$$(T_j + 1)V \;\subseteq\; E^+_j(V).$$

Proof. The Hecke quadratic relation $(T_j - q)(T_j + 1) = 0$ gives $T_j(T_j + 1) = q(T_j + 1)$ in $H_q(S_n)$. Hence for any $x \in V$, $T_j\,(T_j + 1)x = q\,(T_j + 1)x$, i.e., $(T_j + 1)x \in E^+_j(V)$. □

2.2 The May-19 adjacent-disjointness lemma

Lemma 5 (May-19 adjacent disjointness)**.** *For every $H_q(S_n)$-module V and every $1 \leq j \leq n-2$,*

$$E_j^+(V) \cap E_{j+1}^-(V) \;=\; 0, \qquad E_j^-(V) \cap E_{j+1}^+(V) \;=\; 0.$$

Proof. This is `2026-05-19-phase-a-general.tex` Lemma 3.3 (the dim $E_j^+ \cap E_{j+1}^-$ direction), plus its symmetrically swapped form. Both are proved by restricting to the subalgebra $\langle T_j, T_{j+1}\rangle$, a quotient of $H_q(S_3)$, and checking the three irreducibles:

- **Trivial** (T_j, T_{j+1} both act as q): $E_j^- = E_{j+1}^- = 0$, so both intersections are trivially 0.
- **Sign** (both act as -1): $E_j^+ = E_{j+1}^+ = 0$, both intersections trivially 0.
- **Standard** (2-dim; in a basis where T_j is diagonal $\mathrm{diag}(q, -1)$, T_{j+1} acts as a Hoefsmit 2-block at axial distance d_0 with $|d_0| \geq 2$, hence non-degenerate with non-zero off-diagonal entries). E_j^+ is spanned by the first basis vector v_a; E_j^- by the second v_b. The eigenvectors of T_{j+1} are non-trivial linear combinations of v_a, v_b (non-degeneracy). So $v_a \notin E_{j+1}^-$ (else $T_{j+1}v_a = -v_a$, requiring the off-diagonal entry $b \mapsto a$ to vanish), and $v_b \notin E_{j+1}^+$ (similarly). Hence both intersections are 0.

The lemma applies to all V since any $\langle T_j, T_{j+1}\rangle$- submodule decomposes into the three irreducibles above. □

2.3 Synthesis: $(T_j + 1)$ is injective on E_{j+1}^+

Proposition 6 (Adjacent injectivity)**.** *For every $H_q(S_n)$-module V and every $1 \leq j \leq n-2$, the restricted map*

$$(T_j + 1)\big|_{E_{j+1}^+(V)} : E_{j+1}^+(V) \;\to\; E_j^+(V)$$

is well-defined (Lemma 4) and injective *(Lemma 5).*

Proof. The image lies in $E_j^+(V)$ by Lemma 4. The kernel of $T_j + 1$ on V is the (-1)-eigenspace $E_j^-(V)$ (since $T_j = -1$ on E_j^- makes $T_j + 1 = 0$ there, and on E_j^+, $T_j + 1 = q + 1 \neq 0$). Hence the kernel of the restricted map is $E_j^-(V) \cap E_{j+1}^+(V)$, which is 0 by Lemma 5. □

3 Outer-factor injectivity and the lower bound

Theorem 7 (Outer-chain injectivity)**.** *For $\lambda = (3, 3, 1^q)$ with $q \geq 4$, the operator*

$$\mathcal{P} \;:=\; (T_{n-4}{+}1)(T_{n-3}{+}1)(T_{n-2}{+}1)$$

acts injectively *on $E_{n-1}^+|_{V_\lambda}$, with image in $E_{n-4}^+|_{V_\lambda}$.*

Proof. By iterated application of Proposition 6, the three restricted maps

$$(T_{n-2}{+}1)\big|_{E_{n-1}^+} : E_{n-1}^+ \to E_{n-2}^+, \qquad (T_{n-3}{+}1)\big|_{E_{n-2}^+} : E_{n-2}^+ \to E_{n-3}^+, \qquad (T_{n-4}{+}1)\big|_{E_{n-3}^+} : E_{n-3}^+ \to E_{n-4}^+$$

are all well-defined and injective. The composition $(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)\big|_{E_{n-1}^+}$ is therefore well-defined as a map $E_{n-1}^+ \to E_{n-4}^+$ and injective as a composition of injections. □

Remark 8. A subtle point: the operator $\mathcal{P}$ is defined on all of V_λ, not just E^+_{n-1}. The statement of Theorem 7 is about the RESTRICTION of $\mathcal{P}$ to E^+_{n-1}. The proof works because each intermediate image $(T_{n-2}+1)\, E^+_{n-1} \subseteq E^+_{n-2}$ stays inside the domain of the next factor's "adjacent injectivity" statement.

Proof of Theorem 3. The upper bound $\dim B^{(\lambda)} \le 2$ is the SRD-331 result (§1). For the lower bound: by Theorem 1,

$$B^{(\lambda)} \;=\; \mathcal{P}\,\Phi_*(B^{(\mu_1)}).$$

$\Phi_*(B^{(\mu_1)})$ is a 2-dimensional subspace of $E^+_{n-1}|_{V_\lambda}$ (2-dim by Theorem 2 and Φ_* injectivity; contained in E^+_{n-1} by construction of Φ_*). By Theorem 7, $\mathcal{P}$ acts injectively on this 2-dimensional subspace. Hence

$$\dim B^{(\lambda)} \;=\; \dim \mathcal{P}\big(\Phi_*(B^{(\mu_1)})\big) \;=\; \dim \Phi_*(B^{(\mu_1)}) \;=\; 2.$$

□

4 Generalisation to other r=1 shapes

The proof of Theorem 3 uses three shape-sensitive inputs:

1. The SRD-style reduction formula (Theorem 1) relating $B^{(\lambda)}$ to $\mathcal{P}\,\Phi_*(B^{(\mu_1)})$. This holds for any $r=1$ rank-zero λ where the same-row-dies conjecture is proved on the same-row part of E^+_{n-1} (SRD framework theorem). For $(3,3,1^q)$ this is SRD-331.
2. Non-degeneracy of the corner-pair $\{c_1, c_*\}$ 2-block ($|d| \ge 2$), which gives Φ_*.
3. The dim of $B^{(\mu_1)}$.

The injectivity argument (Proposition 6 and the chain Theorem 7) is *shape-agnostic*: it uses only the universal adjacent-disjointness lemma and the image-landing lemma, both of which hold in any $H_q(S_n)$-module.

Theorem 9 (General r=1 dim equality)**.** *Let $\lambda \vdash n$ be a rank-zero partition with exactly one length-preserving corner c_1, column-1 bottom c_*, and $|c(c_*) - c(c_1)| \ge 2$. Assume the same-row-dies conjecture (SRD-331 Conjecture 5.1) holds for λ, so the reduction formula (1) applies with $\mathcal{P} = (T_\ell{+}1)(T_{\ell+1}{+}1)\cdots(T_{n-2}{+}1)$ ($n-1-\ell$ factors) and $\mu_1 = \lambda \setminus \{c_1, c_*\}$. Then*

$$\dim B^{(\lambda)}_{\ell+1} V_\lambda \;=\; \dim B^{(\mu_1)}_{\ell(\mu_1)+1} V_{\mu_1}.$$

Proof. The reduction formula gives $\dim B^{(\lambda)} \le \dim B^{(\mu_1)}$ (via Φ_* injective and dimension-non-increase under $\mathcal{P}$). For the reverse: $\Phi_*(B^{(\mu_1)}) \subseteq E^+_{n-1}|_{V_\lambda}$, and the iterated (T_j+1)-chain $(T_\ell+1)\cdots(T_{n-2}+1)$ runs through indices $\ell, \ell+1, \ldots, n-2$ in increasing order. Iteratively applying Proposition 6: $(T_{n-2}+1) : E^+_{n-1} \to E^+_{n-2}$ is injective; then $(T_{n-3}+1) : E^+_{n-2} \to E^+_{n-3}$ is injective; … ; finally $(T_\ell+1) : E^+_{\ell+1} \to E^+_\ell$ is injective. Composition injective. Hence $\dim B^{(\lambda)} = \dim \mathcal{P}\,\Phi_*(B^{(\mu_1)}) = \dim B^{(\mu_1)}$. □

Remark 10 (Application range)**.** Theorem 9 immediately gives $\dim B$ in closed form for any $r=1$ shape where SRD-style same-row-dies has been established and the μ_1-dim is known. Currently the SRD framework is established at:

- Hooks $(k, 1^{n-k})$, with $\mu_1 = (k-2, 1^{n-k})$. Inductive on k. (Closed by the Shift Lemma + hook j-formula chain.)
- $(3,3,1^q)$, with $\mu_1 = (3,2,1^{q-1})$. SRD-331.

- $(4, 2, 1^q)$, with $\mu_1 = (4, 1^q)$ via SRD-421 same-row analysis. The j-formula in this case is the hook $(4, 1^q)$ formula, $\dim = 1$ (hook dim-1, May-13).
- In general, $(k, 2, 1^q)$ for various k (recursive).

5 Computational verification

Mod-P verification at $P = 100\,003$, $q = 11$ (the seed specialisation), for $\lambda = (3, 3, 1^q)$ at $q \in \{4, 5, 6\}$:

- Adjacent-disjointness $E_j^- \cap E_{j+1}^+ = 0$ at $j \in \{n-4, n-3, n-2\}$: verified at each q.
- The outer-chain rank: $\operatorname{rank}(\mathcal{P}|_{E_{n-1}^+}) = \dim E_{n-1}^+|_{V_\lambda}$ at each q. Specifically:

q	n	$\dim E_{n-1}^+\|_{V_\lambda}$
4	10	85
5	11	133
6	12	196

- Direct dim: $\dim B^{(\lambda)} = 2$ at each q via direct computation of $R'_{\ell+1} \cdots R'_n V_\lambda$.

  Scripts: `~/projects/scratch/2026-05-13-dim2-331-lower/`.

6 Honest status

What we proved. $\dim B^{(\lambda)} = 2$ structurally for $\lambda = (3, 3, 1^q)$ at every $q \geq 4$, closing the lower-bound gap of SRD-331 Remark 4.6.

What we generalised. An r=1 dim-equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$ (Theorem 9) under the SRD-style hypothesis. This is shape-agnostic and uses only universal Hecke-algebra facts (image landing + adjacent disjointness).

What this clarifies. The earlier "stuck" position analysis in `2026-05-13-dim2-331/proof-sketch.md` (the natural attempt: position-of-$(n-2)$ separator preserved by outer factors; fails because T_{n-3}, T_{n-2} move letter $n-2$) was looking in the wrong direction. The correct argument doesn't track positions at all — it uses the abstract $H_q(S_n)$-module-structural fact that adjacent T_j-eigenspaces of opposite sign intersect trivially.

What this does NOT generalise to. Multi-corner ($r \geq 2$) shapes. There, the reduction formula has multiple Φ_X-branches, and the lower bound requires a separation argument between branches (see May-12 evening for $r = 2$ at $(3, 2, 1^*)$). The adjacent- injectivity argument shows that each individual $\Phi_X(B^{(\mu_X)})$- branch is preserved by the outer chain, but the r branches may *collide* after the outer factors. So $\dim B^{(\lambda)} \leq r \cdot \dim B^{(\mu_X)}$, not equal in general. Separation between branches needs a different tool (position-of-n, May-12 evening template).

Status of related conjectures.

- For $(3, 3, 1^q)$, $q \in \{2, 3\}$: same-row-dies holds (SRD-331); $\dim B^{(\mu_1)} = 2$ at $q \in \{2, 3\}$ is computational (May-12 evening rigorous at $q \geq 4$). The current theorem is unconditional at $q \geq 4$; at $q \in \{2, 3\}$ it gives $\dim B^{(\lambda)} = 2$ *conditional on the computational μ_1-dim.*

- For the $(4,3,1^q)$ family ($r=2$, conjecturally $\dim = 5$): the present paper closes one ingredient but *not* the full lower bound; see Remark 11 below.

Remark 11 (Partial application to $(4,3,1^q)$)**.** In the $r=2$ reduction formula for $\lambda = (4,3,1^q)$ (`2026-05-13-4-3-1n-lower-bound-analysis.tex`), $B^{(\lambda)} = \mathcal{O}_\lambda[\Phi_A^\lambda(B^{(\mu_A)}) + \Phi_B^\lambda(B^{(\mu_B)})]$, the present paper provides one ingredient: $\mathcal{O}_\lambda$ acts injectively on each individual $\Phi_X(B^{(\mu_X)})$ (each lies in E_{n-1}^+ and adjacent-injectivity propagates). Hence the lifted inner bases $\{F_A^1, F_A^2\}$ and $\{F_B^1, F_B^2, F_B^3\}$ are linearly independent within each Φ_X-class.

However, this is *not* sufficient to close the full 5-vector independence. The lower-bound-analysis paper's Theorem 3.2 reduces the problem to projected versions (a) $\{\pi_n^{(c_1)} F_A^i\}$ and (b) $\{\pi_n^{(c_2)} F_B^j\}$. The projected vectors are not in E_{n-1}^+ (the seminormal projection breaks the $+q$-eigenvector structure), so adjacent-injectivity does not directly apply — a cross-class linear combination could carry c_3-overlap invisible to either projection alone. The honest status of the $(4,3,1^q)$ lower bound is unchanged: tracer-pair Hoefsmit remains the open ingredient.

7 What's next

Immediate corollaries (not in this paper). The adjacent-injectivity framework gives an immediate dim-equality $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$ for any r=1 shape with SRD-style same-row-dies. This subsumes:

- Hooks $(k, 1^{n-k})$ at $k \geq 3$ ($\mu_1 = (k-2, 1^{n-k})$): $\dim B^{(\lambda)} = \dim B^{(\mu_1)}$. For hooks, $\dim B^{(\mu_1)}$ is given by hook-dim-1 paper (May-13). Recursive descent gives $\dim B^{(\lambda)} = 1$ for all hooks.
- $(k, 2, 1^q)$ via SRD-421 framework: same-row-dies on row 1, $\mu_1 = (k, 1^{q+1})$. Reduces to hook μ_1.
- $(k, k, 1^q)$ at $k = 3$ (this paper).

Deeper question. The adjacent-injectivity argument suggests a structural fact: the operator $\mathcal{P} = (T_{n-a}+1)\cdots(T_{n-2}+1)$ (for any number of factors) is *injective* on $E_{n-1}^+|_V$ for any $H_q(S_n)$-module V. This is a clean general fact independent of the rank-zero programme; it says that the iterated $+q$-eigenspace descent through adjacent indices loses no dimension. The proof is the straightforward chain of adjacent-disjointness arguments.

This in turn suggests that for any reduction formula of the SRD-style, $\dim B^{(\lambda)}$ is determined entirely by $\dim B^{(\mu_1)}$ plus the dim of the same-row-dies part (if non-zero). For $r=1$ shapes with the same-row part dying, this gives the cleanest possible reduction.